

KALIANGET MADURA ISLAND, IndonesiaWMO: **969730**Lat: **7.0413S**Lon: **113.9156E**Elev: **3**StdP: **101.29**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.7	24.1	19.8	14.5	27.3	20.5	15.2	27.3	14.6	28.3	12.9	28.8	1.0	300	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	6.0	33.4	27.4	32.9	27.3	32.4	27.2	28.9	32.1	28.4	31.7	28.0	31.4	4.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.2	31.7	27.5	23.4	31.2	27.0	22.7	30.8	94.3	32.2	91.8	31.8	89.9	31.6	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.8	7.8	19.6	34.4	3.6	0.5	17.0	34.7	14.9	35.0	12.9	35.3	10.3	35.7		
(5)			18.4	29.5	2.5	0.7	16.6	30.0	15.1	30.4	13.7	30.9	11.9	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	27.9	27.9	28.2	28.5	28.6	28.0	27.6	27.9	28.4	29.4	29.3	28.4		
	DBStd	1.04	0.92	0.88	0.79	0.81	0.92	0.92	0.85	0.72	0.78	0.85	1.16	1.02		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6691	553	501	564	553	575	540	547	554	553	601	579	570		
	CDD18.3	3649	295	268	306	303	317	290	289	296	303	342	329	312		
	CDH23.3	41138	3063	2780	3283	3433	3692	3224	3084	3185	3508	4305	4131	3451		
	CDH26.7	14429	904	831	1083	1213	1367	1038	883	931	1276	1892	1820	1192		
(14) Wind	WSAvg	2.9	3.2	2.7	1.8	1.9	2.6	3.0	3.7	4.3	3.9	3.4	2.3	2.1		
(15) Precipitation	PrecAvg	1497	234	213	228	143	152	73	30	10	16	41	118	234		
	PrecMax	2199	445	390	527	298	633	322	301	84	103	241	374	517		
	PrecMin	738	13	76	44	22	2	0	0	0	0	0	0	49		
	PrecStd	351	106	82	108	79	137	84	61	21	29	62	95	118		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	32.6	32.8	32.7	32.7	32.1	31.4	31.6	32.7	34.0	34.2	33.5		
		MCWB	27.1	27.8	27.7	27.7	27.5	27.0	26.0	26.1	26.9	27.1	27.9	27.5		
	2%	DB	31.8	31.8	32.2	32.1	32.1	31.5	30.9	31.0	32.1	33.2	33.5	32.5		
		MCWB	27.1	27.5	27.5	27.5	27.3	26.7	25.9	25.8	26.3	27.0	27.6	27.6		
	5%	DB	31.1	31.2	31.6	31.6	31.7	31.0	30.5	30.6	31.6	32.7	33.0	31.9		
		MCWB	27.0	27.0	27.3	27.3	27.1	26.5	25.7	25.5	26.0	26.9	27.4	27.4		
	10%	DB	30.4	30.5	31.0	31.1	31.1	30.4	30.0	30.2	31.1	32.2	32.4	31.1		
		MCWB	26.7	26.7	27.1	27.2	26.8	26.2	25.5	25.2	25.8	26.8	27.2	27.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.7	28.9	28.7	28.7	28.5	28.1	27.7	27.6	28.7	29.6	29.6	29.0		
		MCDB	30.9	31.7	31.8	31.8	31.5	30.9	30.1	30.7	31.9	32.9	33.3	32.3		
	2%	WB	27.7	27.9	28.1	28.1	28.0	27.4	26.8	26.8	27.7	28.6	28.7	28.2		
		MCDB	30.6	31.1	31.4	31.3	31.3	30.5	29.8	30.2	31.3	32.2	32.5	31.7		
	5%	WB	27.2	27.3	27.6	27.7	27.6	27.0	26.4	26.2	27.0	27.9	28.0	27.7		
		MCDB	30.3	30.4	30.8	30.8	30.8	30.2	29.4	29.7	30.7	31.7	31.9	31.1		
	10%	WB	26.7	26.9	27.2	27.3	27.2	26.6	26.0	25.8	26.4	27.2	27.5	27.2		
		MCDB	29.7	29.8	30.2	30.3	30.3	29.6	29.1	29.2	29.9	31.0	31.3	30.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.6	5.9	5.5	5.2	5.0	5.0	5.0	5.8	6.0	5.8	5.6		
		MCDBR	6.2	6.4	6.6	6.0	5.6	5.4	5.4	5.3	6.3	6.5	6.4	6.4		
	5% WB	MCWBR	3.0	3.1	3.3	2.9	2.6	2.5	2.5	2.6	3.0	2.8	2.8	3.1		
		MCWBR	5.8	6.1	6.2	5.8	5.5	5.1	5.2	5.0	5.9	6.0	6.0	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.514	0.512	0.483	0.453	0.433	0.421	0.412	0.427	0.439	0.483	0.507	0.522		
		taud	2.173	2.184	2.266	2.359	2.423	2.458	2.473	2.405	2.352	2.228	2.177	2.141		
	Ebn at Noon		835	837	848	847	836	832	848	858	873	850	835	825		
		Edh at Noon	158	157	143	125	112	105	105	117	129	149	157	162		
(44) All-Sky Solar Radiation	RadAvg		4.93	5.21	5.39	5.37	5.17	5.03	5.37	6.00	6.53	6.43	5.72	4.75		
	RadStd		0.48	0.41	0.42	0.45	0.38	0.33	0.28	0.13	0.25	0.51	0.55	0.49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.24	+0.73	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page